

FrontierAtlas Global Intelligence Graph Architecture

Autonomous Scalable Ingestion, Multi-Tier LLM Orchestration, & Deterministic Entity Resolution | 500k+ Scale Design

1. Executive Overview & System Topology

FrontierAtlas continuously ingests, structures, resolves, and enriches multi-dimensional signals across the AI venture ecosystem. The architecture reconciles the throughput of traditional web scrapers with the structured extraction power of LLMs, enforcing strict zero-hallucination policies, sub-24-hour freshness boundaries, and sub-millisecond graph queryability.

Ingestion Sources: <i>arXiv, PapersWithCode, YC, Hugging Face, 5 AI News Feeds, 5 AI Job Boards, GitHub</i>	Crawler Mesh: <i>Async aiohttp + Playwright Stealth + User-Agent Pool + Token Bucket Limiter</i>
Resilience Engine: <i>DOM Pruning (413 Immunity) + Gemini Flash / Groq Llama 3 / Heuristic Fallback</i>	Resolution & Storage: <i>Deterministic Suffix Stripping + RapidFuzz + PostgreSQL / Neo4j / Qdrant</i>

2. Scale Strategy: Harvesting 500,000+ Entities Autonomously

To scale to 500,000+ startups, products, research papers, and live market signals without human intervention, the system deploys a distributed orchestration mesh:

- **Distributed URL Frontier & Workflows:** Orchestrated via **Temporal.io** and **Celery on Redis/RabbitMQ**. Implements partitioned URL priority queues with per-domain rate limiting, ensuring polite concurrent crawling without triggering IP bans.
- **Autoscaling Worker Fleet:** Kubernetes Event-driven Autoscaling (KEDA) dynamically scales crawler worker pods from 5 to 150 instances based on queue depth and endpoint responsiveness.
- **High-Throughput Research Harvesting:** Bulk OAI-PMH harvest protocol for arXiv combined with PapersWithCode open metadata dumps to process hundreds of thousands of papers into partitioned S3 Parquet datasets in minutes.
- **Dynamic GitHub Metrics Tracking:** Pooled authenticated GitHub GraphQL workers with unauthenticated HTML regex scrapers resolve live stargazers, forks, and commit velocities across 100k+ repositories with continuous caching.

3. Resilient LLM Orchestration: Managing 413s & 429s

LLM pipelines in production frequently fail due to payload size overflows (HTTP 413) and aggressive rate limits (HTTP 429). FrontierAtlas guarantees zero-downtime execution via a multi-tier defense:

A. 413 Payload Too Large Elimination (Semantic Chunking & DOM Pruning)

- **AST / DOM Tree Pruning:** Strips non-semantic elements (<script>, <style>, <svg>, <iframe>, base64 images, navbars, footers).
- **Semantic Density Condenser:** Translates cleaned DOM structures into dense markdown representations preserving headings, tables, and JSON-LD schema.
- **Sliding Window with Overlap:** Enforces a strict 3,500-token chunk window with 250-token overlap, ensuring payloads never breach provider context limits.

B. 429 Rate Limit Mitigation & Multi-Tier Fallback Chain

- **Token Bucket Rate Limiting:** In-memory and Redis token buckets enforce client-side RPM and TPM ceilings per API provider.
- **Full Jitter Exponential Backoff:** $Delay = uniform(0.1, \min(Cap, Base \times 2^{attempt}))$. Eliminates thundering-herd retry storms across workers while parsing explicit Retry-After headers.
- **Multi-Tier Circuit Breaker Chain:**
 - *Tier 1 (Primary):* Gemini 1.5 / 2.0 Flash (Fast JSON mode, high context, cost-efficient).
 - *Tier 2 (Failover):* Groq Llama 3.3 70B / 8B (Sub-second low-latency inference).
 - *Tier 3 (Local Fail-Safe):* Deterministic Rule-Based Heuristics (Guarantees 100% uptime during global API outages).

4. Freshness Tracking: 24-Hour Guarantee & Zero-Duplicate Crawling

Extreme freshness is enforced across 5 AI news feeds and 5 AI job boards to guarantee market intelligence reflects events in the preceding 24-hour window:

- **Distributed Bloom Filter Deduplication:** Redis Scalable Bloom Filters store SHA-256 signatures of (entity_id + source_url) with < 0.001% false positive rate.
- **Content Hash Deltas:** Computes cryptographic hashes of article bodies to detect content alterations and avoid redundant processing.
- **HTTP Conditional Requests:** Automates ETag and If-Modified-Since headers to bypass un-updated sites via HTTP 304 Not Modified.
- **Precision Date Normalization:** Custom parsing pipeline resolving ISO-8601, RFC-2822, UNIX epoch, JSON-LD microdata, and relative human timestamps ('2 hours ago', 'yesterday') with strict 24.0-hour boundary enforcement.

5. Deterministic Entity Resolution Engine

Normalizes noisy, heterogeneous entity representations (e.g. 'OpenAI, Inc.', 'Open AI', 'openai.com') into canonical nodes in the Intelligence Graph:

- **Rule-Based Normalizer:** Strips legal suffixes (Inc, LLC, Ltd, Corp, PBC, GmbH, SAS, Pte Ltd, Labs, Technologies) and extraneous punctuation.
- **Domain & Hostname Matching:** Maps root domain URLs to canonical startup organizations with 0.95+ confidence.
- **Hybrid Fuzzy Matching:** RapidFuzz composite scoring combining Token Sort Ratio and Jaro-Winkler similarity against a 50+ entity seed database (Threshold >= 0.85).
- **Immutable Audit Trail:** Generates Entity Mapping Logs recording raw_name, canonical_name, confidence_score, method_used, and source_url.

6. Anti-Bot Navigation & Stealth Strategy

Bypasses Cloudflare Turnstile, Datadome, and Akamai on high-value intelligence sources via layered evasion techniques:

- **Browser Fingerprint Realism:** Rotates realistic User-Agents matching Client Hints (sec-ch-ua, sec-ch-ua-platform, Sec-Fetch-*).
- **Playwright Async Stealth:** Evasions patching navigator.webdriver, WebGL parameters, canvas noise, and human-like bezier mouse paths for JavaScript SPAs.
- **Residential Proxy Mesh:** Domain-pinned sticky IP rotation through residential backbones.

7. Polyglot Storage Architecture & Deliverables Summary

Relational / Events	PostgreSQL + TimescaleDB	Canonical entity metadata, time-series star metrics, ACID logs
Knowledge Graph	Neo4j / Memgraph	Multi-dimensional graph edges (Startup -> Product -> Paper -> Repo)
Vector Store	Qdrant / pgvector	Dense semantic embeddings for hybrid search & discovery
Data Lakehouse	Parquet on S3 / Iceberg	Petabyte-scale historical analytics and cold snapshot archive

Trial Deliverables Summary:

- ✓ **6 Output Tabs:** Startups (1,000+), Products (1,000+), Research Papers (1,000+ with GitHub stars), Jobs (24h fresh), News (24h fresh full-text), Entity Mapping Log.
- ✓ **Engineering Codebase:** Complete async pipeline CLI, unit test suite, and automated Google Sheets / Excel / JSON export engines.